

FINAL EXPLANATION: BIOLOGY GRADE 11

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 1.3 TAXONOMY II: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How can we identify and classify living organisms accurately? You will write a Claim-Evidence-Reasoning (CER) Final Explanation using what you observed in Lessons 1 to 8 about the organisms around our school: maize, bean plant, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard and goat. Answer each section in the space below it. Use evidence from your own observations, your group work and your Driving Question Board. Use your own words and record scientific terms in English. You may discuss ideas in Kiswahili or your mother tongue before you write. Then use your answers to help your group build its Local Biodiversity Classification Guide.

Part 1: The problem with our own groupings

In Lesson 1 your group sorted the organisms from around the school and compared your grouping with other groups in a gallery walk. (a) What rules did different groups use, and what did you notice when you compared the groupings? (b) Explain why a standard system of classification is needed so that any person anywhere can identify and study an organism.

(a) Different groups sorted the organisms by size, colour, habitat or usefulness as food. When we compared the groupings in the gallery walk they did not agree, because every group chose different rules, so no single grouping worked for everyone. (b) A standard system is needed because personal rules give different results for different people. If everyone uses the same shared, observable characteristics, any person anywhere can identify and study an organism and get the same result. This lets scientists organise and communicate about living things accurately.

Part 2: Classifying plants

Use the plants you examined (moss, fern, maize and bean plants). (a) Name the three plant groups you studied and state the observable features that place a plant in each group. (b) Explain why grouping plants by these features is more reliable than grouping them by size,

(a) Bryophyta (moss): small, rhizoids instead of true roots, no vascular tissue, spores in capsules, grows in damp places. Pteridophyta (fern): true roots, a rhizome, fronds with brown sori on the underside, vascular tissue, reproduces by spores, no seeds or flowers. Spermatophyta (maize and bean): roots, stems and leaves with vascular tissue, reproduce by seeds; maize and beans produce flowers and seeds (grains and pods), and can be divided into monocots and dicots. (b) These features can be observed and checked by anyone, so every group places a plant in the same division. Size, colour or use as food depend on personal opinion, so groups disagreed, and moss, for example, was wrongly placed with grass or 'small green things'.

colour or use as food.	
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Part 3: Classifying animals

Think about the animals in our collection (earthworm, snail, mosquito, butterfly, chicken, lizard, goat). (a) Choose three animals from different phyla and write a CER statement placing each in its phylum, using observable body features. (b) Explain how the goat, chicken and lizard are further separated within their phylum.	(a) Claim: the earthworm is in Annelida, the snail is in Mollusca and the mosquito is in Arthropoda. Evidence: the earthworm has a soft segmented body with no legs; the snail has a soft body with a shell; the mosquito has jointed legs and a hard outer covering (exoskeleton). Reasoning: phyla are decided by shared, checkable body features such as segmentation, a shell, jointed legs with an exoskeleton, or a backbone, not by size, colour or habitat, so anyone examining the animal reaches the same phylum. (b) The goat, chicken and lizard are all Chordata because they have a backbone. Phyla are divided into classes using more specific features such as body covering, limbs, respiration and reproduction: the goat has hair (Mammalia), the chicken has feathers (Aves) and the lizard has scales (Reptilia). Arthropoda are likewise divided into classes, for example Insecta, where the mosquito and butterfly belong.
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Part 4: Dichotomous keys

(a) Describe how a dichotomous key is built and used. (b) Write the first two couplets of a key you could use to separate three organisms from our collection. (c) Explain why different groups using the same key reach the same answer, unlike your Lesson 1 groupings.	(a) A dichotomous key is a set of numbered couplets. Each couplet gives two contrasting, observable choices; you choose the one that fits the organism and follow it to the next couplet until you reach one name. (b) Example for moss, fern and maize: 1a. Plant has no true roots or vascular tissue: moss (Bryophyta). 1b. Plant has true roots and vascular tissue: go to 2. 2a. Reproduces by spores on the underside of fronds, no seeds: fern (Pteridophyta). 2b. Reproduces by seeds: maize (Spermatophyta). (c) The key uses clear, observable features, not personal opinions such as size, colour or usefulness. Each step has only two choices, so different people following the same couplets get the same identification, whereas our own groupings depended on the rules each group invented.
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Part 5: Viruses and the final explanation

(a) What are the characteristics of viruses, and give two examples of why they are important in agriculture, health or biotechnology. (b) Now write your Final Explanation for the driving question 'How can we identify and classify living organisms accurately?' Use a claim, evidence from at least two lessons, and reasoning.	(a) Viruses are non-cellular, have a protein coat and genetic material, and reproduce only inside host cells, so they differ from cellular organisms. They matter in agriculture because maize lethal necrosis disease harms Kenyan maize and cassava mosaic disease harms cassava; in health because viruses cause HIV, measles, polio and COVID-19, and vaccines protect against some of them; and viruses are also used in biotechnology. (b) Claim: Living organisms can be identified and classified accurately by using a standard system based on shared, observable characteristics, together with tools such as dichotomous keys. Evidence: In Lesson 1 our groupings of the school organisms by size, colour or usefulness differed from group to group. Later we placed plants into Bryophyta, Pteridophyta and Spermatophyta using features like rhizoids, sori on fern fronds and seeds, and placed animals into phyla such as Chordata and Arthropoda and then into classes using features like backbone, jointed legs, and hair, feathers or scales. When different groups used dichotomous keys they reached the same answer. Reasoning: Because these features can be checked by anyone, the same organism is placed in the same group by everyone, so anyone anywhere can study and communicate about biodiversity. This replaces our Lesson 1 groupings, which varied between groups. Viruses show that
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Say how your thinking has changed since your Lesson 1 model.	classification depends on structure and characteristics, since they are non-cellular. A remaining question is how classification knowledge can be applied outside the classroom.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim answers the driving question	Gives a clear claim that a standard system based on shared, observable characteristics and dichotomous keys allows accurate identification and classification.	Gives a claim about classification but it is general or only partly linked to observable characteristics and keys.	Claim is missing, unclear or does not answer the driving question.
Evidence from the lessons	Uses specific evidence from several lessons, such as the school organisms, moss, fern, maize, bean, earthworm, snail, mosquito, goat, chicken and lizard, and the key results.	Uses some correct evidence from the lessons but it is limited or lacks detail.	Evidence is vague, incorrect or not taken from the lessons.
Scientific reasoning and use of terms	Explains why observable shared features give the same result for everyone while size, colour or use do not, and uses divisions, phyla, classes, couplets and viruses correctly.	Gives some reasoning and uses most terms correctly.	Little or no reasoning; terms are missing or misused.
Dichotomous key and virus knowledge	Writes correct contrasting couplets and accurately describes virus characteristics and their importance with Kenyan examples.	Writes a mostly workable key and gives some correct virus facts.	Key is not workable and virus ideas are missing or incorrect.
Revision of thinking	Clearly compares the Lesson 1 groupings with the final model and states a relevant new question.	Mentions that thinking changed but with limited comparison.	Does not show how thinking has changed.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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